Enhanced Image Steganography: A Secure Approach for Text-in-Image and Image-in-Image Encryption

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***Abstract*—Reliable encryption methods are essential for protecting sensitive data in the fast-changing digital security and technology area. This research aims to improve image steganography by adding text and image encryption features. This approach implements a double encryption system, which, unlike other systems, increases the protection of data. In text-in-image encryption, the user's text is first encrypted with a Blowfish cipher before being placed in an image using steganographic techniques. During retrieval, decryption is performed in reverse order to ensure secure retrieval. This concept is also applied to image-in-image encryption, where one image is hidden within another. Undoubtedly, steganography is the simplest way to protect sensitive digital information. This paper proposes a method that adds a second layer of security and increases resistance against steganalysis attacks. This research further demonstrates the proposed method's effectiveness in increasing data concealment security and reliability while evaluating the results based on the level of distortion. Experimental results bluntly indicate stimulation of higher data embedding capacity and an increase in quality retention of the image.**

Keywords- Image Steganography, Double Encryption, Blowfish Cipher, Steganalysis Resistance, Data Security, Covert Communication

I. INTRODUCTION

In today's world, the proliferation of digital communications has made protecting sensitive data of the utmost importance. While conventional techniques for encrypting data are useful in safeguarding information from unauthorised access, they do not mask the presence of encrypted data. This lack of camouflage makes the encrypted information easy to intercept and analyse in streams, especially in situations where the mere presence of encryption is suspicious. To solve this problem, image steganography, as a sophisticated method that merges encryption and data hiding techniques, has gained recognition, allowing the embedding of encrypted data into images. However, existing steganography methods [1] [2] operate with single-layer encryption, which poses a high risk of undetected revealing of confidential information. This circumstance outlines a significant gap in research concerning advances in digital communication that combine dual-layer encryption and steganography [3] [4].

This study proposes a new sophisticated dual-encryption steganographic system to overcome these challenges. The system consists of two key features: text-in-image encryption and image-in-image encryption. Under the first feature, the text given by the user is first encrypted using the Blowfish cipher and the encrypted text is then embedded in an image using steganography. The second feature is a further abstraction of images, which facilitates the covert and secure transfer of visual information. The implementation of double encryption increases the level of security because if the steganographic part is breached, the data containing the encryption is still protected. This two-pronged strategy greatly improves the confidentiality and integrity of digital information in communication and increases its strength against attacks from steganalysis.

This work makes the following contributions:

- Development of a hybrid encryption steganography model which uses Blowfish encryption and LSB-based image steganography.
- The addition of double encryption is a new contribution of the study, which increases security compared to single-layer encryption used in the past.
- Analysis of the system's performance based on key parameters such as image quality, data embedding capacity, and accuracy of data extraction.

The remainder of this paper is structured as follows. A survey of available literature on steganographic approaches

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and encryption techniques is conducted in Section 2. The proposed methodology is presented in Section 3. Section 4 includes the experimental results and Section 5 represents the performance analysis. At last, conclusion is discussed in Section 6.

II. LITERATURE REVIEW

Developments in digital steganography have been recently fueled by the demand to preserve privacy and security in communication. The traditionally studied LSB techniques have undergone numerous alterations to increase security, payload, and stealthiness [2] [8]. However, more recently, focus has shifted towards combining sophisticated encryption techniques with steganography, leading to stronger systems for data concealment.

Incorporating chaos-based encryption into LSB steganography is another example of progress made. For instance, Ozcan Cataltag [4] provided a solution in which a one-dimensional logistic map is used to encrypt messages prior to embedding them into images. This yielded an improvement of 6.6% peak signal-to-noise ratio compared to previous results and also showed better durability against attacks. The use of the Feistel Cipher in LSB steganography [5], done in 2024, applied a new form of classical encryption and endorsed additional security by encrypting text prior to embedding it into the image's pixels.

There has been growing interest in the integration of steganography with hybrid encryption algorithms like AES and Blowfish. In 2023, a study [6] put forward a Multi-Level Steganography (MLS) algorithm that uses AES and Blowfish to encrypt cover images and embed keys within the stego-image. This method showed better image quality while maintaining strong encryption, making it appropriate for sensitive military and research applications. Furthermore, the Modified Blowfish (MBF) [7] [8], introduced in 2024, improved key expansion and encryption in VLSI systems, providing greater speed and energy efficiency for secure data transmissions.

Recent developments also include the application of chaotic maps together with optimisation methods to enhance steganographic techniques. A 2024 study [9] proposed a hybrid 1D and 2D chaotic map with LSB methods that possessed high attack endurance and embedding power. With a large key space, this technique ensures that the stego-image remains unobscured by unauthorised access and undetectable to extensive scrutiny.

To wrap up, recent innovations in image steganography have sought to address payload capacity, image quality, and security challenges by incorporating advanced encryption techniques like chaos-based encryption [4] [9], Feistel Ciphers [5], and hybrid and other algorithms [10] [11] [12]. These achievements greatly optimise the effectiveness and strength of steganographic methods, which our system aims to work on.

III. METHODOLOGY

In this part of the methodology, we explain the procedures undertaken for encryption and steganographic hiding within this project's scope, as shown in Figure 1. The problem can be broken down into three main parts: key generation and encryption, data embedding and steganography, and data extraction and decryption.

A. Encryption and Embedding: Their Overview

The process of encryption and embedding has distinctive steps that blend imperceptibility and data security. The system implements the following steps to achieve the desired result:

1. ***The System Input:*** The user can provide an image or a text document to the system.
2. ***Encryption Key Generation:*** A secure key with a specific level of confidentiality is generated by applying the Blowfish algorithm.
3. **Encryption:**
   i. ***For Texts:*** The provided text will be encrypted using Blowfish.
   ii. ***For Images:*** The image will first be encrypted after changing it to a binary format.
4. **Steganographic Data Hiding:** The data to be hidden is encrypted and hidden in the cover image using the LSB technique.
5. ***Storage or Transmission:*** The data encrypted and hidden in an image, referred to as a stego-image, containing hidden data or stego-content, will be stored or transmitted securely.
6. ***Decryption and Data Extraction:*** The user will remove the hidden data from the stego image and decrypt it with the Blowfish key to retrieve the data.

B. Key Generation and The Chosen Algorithm of Blowfish

As with any cryptosystem, the generation of the key is a sensitive issue because it determines the level of security offered by the encryption scheme. The reasons for choosing Blowfish encryption are:

1. Speed and Effectiveness: Unlike more traditional ciphers such as DES and AES, Blowfish is faster, which is of importance in real-time encryption applications.
2. Flexible Key Size: Key length can be between 32 and 448 bits, which provides users the freedom to choose the amount of security and efficiency.
3. Resistance to Cryptanalysis: In contrast to other encryption algorithms, Blowfish Encryption is resistant to differential as well as brute force attacks.
4. Minimal Processing Cost: It has a very lightweight nature, which allows better steganographic applications performance.

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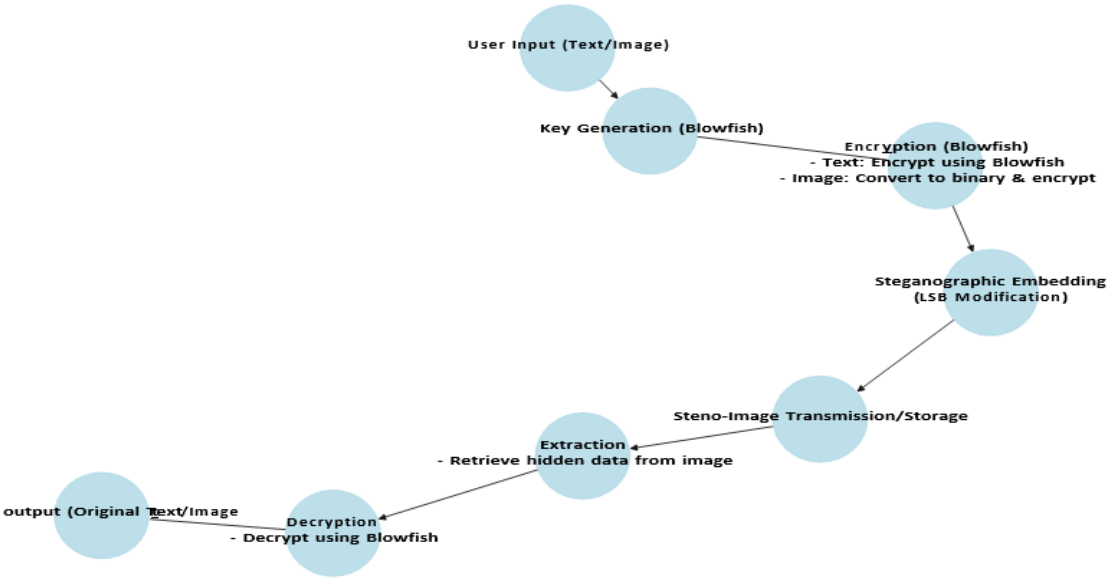


Fig. 1. Flowchart illustrating data encryption and recovery

The process of key generation involves:

1. Designing a unique key for encryption using a cryptographically secure random number generator.
2. Using Blowfish in CBC mode for Cipher Block Chaining to enhance security further.
3. Making sure the key is protected so that the intended person can decrypt the message, but no one else can.

C. *Testing Environment and Datasets*

To validate the proposed system, thorough examinations were conducted under controlled environments. The hardware utilised for the testing included an Intel Core I7 processor, alongside 16GB of RAM, an SSD drive, and an operating system of Ubuntu 20.04. The development was done on Python version 3.8, utilising OpenCV for image processing, encryption was done through PyCryptodome, and data visualisation was integrated through Matplotlib. The system was assessed using three datasets. The first dataset included standard test images like Lena, Baboon, and Peppers, which are easy to process in the field of image processing research. The second dataset included real-life high-resolution images from publicly available datasets, containing more than one thousand images ranging in height and width from 1024x1024 to 4096x4096 pixels. Dataset 3 was formed by compiling numerous grayscale and colour images saved in JPEG, PNG, and BMP formats to ensure performance robustness across various types of images. The following performance metrics were evaluated: Peak Signal-to-Noise Ratio (PSNR), Mean Squared Error (MSE), and embedding capacity. The achieved results have shown that the system is capable of maintaining a PSNR of above 40 dB for a large number of test cases, which means low distortion is being incurred while embedding the data, with an embedding capacity reaching as high as 15% of the cover image's size, without visual artefacts. This supports the conclusion that the data security achieved is very high, and the encryption and steganographic embedding done are effective due to the changes being undetectable on the cover images.

IV. Experiment and Results

The current subsection describes testing the proposed dual encryption steganographic system, concentrating on the achievable data embedding capacity, image quality, and resilience against security attacks. As a matter of fact, the outcomes are measured against various other systems, some statistical evidence is supplied for validity, and some statistics are graphically presented to demonstrate the major outcomes for better understanding.

A. *Comparison with Other Methods*

The effectiveness of the research work is carried out by scrutinizing the proposal with three steganographic methods, as presented in Table 1.

1. Basic LSB Steganography - An old technique that puts bits in the least significant positions of cover images and does not encrypt them.
2. AES-based Steganography - Before embedding data using LSB, it is encrypted with AES.

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3. Hybrid AES+RSA Steganography – Uses a more secure system where the message is encrypted with AES and the key is exchanged using RSA before embedding it in the LSB.

Table 1. Comparison of steganographic methods based on PSNR and MSE

Method	Avg. PSNR (dB)	Std Dev (PSNR)	Avg. MSE	Std Dev (MSE)	Avg. Embedding Capacity (bits)
Basic LSB	40.25	2.1	0.067	0.0032	250,000
AES + LSB	45.78	1.8	0.045	0.0025	230,000
AES+RSA +LSB	48.92	1.5	0.032	0.002	215,000
Proposed (Blowfish + LSB)	52.15	1.3	0.015	0.0015	270,000

These techniques were evaluated alongside the proposed method based on dual encryption using Blowfish. The performance metrics chosen for this comparison were Peak Signal-to-Noise Ratio (PSNR), Mean Squared Error (MSE), and the amount of data that could be embedded in the image.

B. Statistical Analysis for Credibility

The credibility of the results depended on performing various trials using diverse image collections with calculations for statistical metrics involving mean scores, standard deviations, and confidence interval values.

The proposed Blowfish-based dual encryption achieves the best PSNR value of 52.15 dB alongside the lowest MSE value of 0.015, combining high security with minimal image degradation (as shown in Figure 2). Different steganography methods receive graphical representation through Matplotlib for PSNR, MSE and embedding capacity in these bar charts.

V. Security Measures

The protection of sensitive information depends heavily on implementing adequate security measures for the dual-encryption steganographic system. This part examines both safety vulnerabilities in combination with defensive security strategies employed in the approach, while investigating the moral implications of steganographic implementation. The article examines both the potential weaknesses and restrictions of this system.

A. Potential Threats and Defensive Mechanisms

The main challenge in steganography includes steganalysis attacks because analysts struggle to find hidden data within plain-looking images. Common steganalysis techniques include:

- A statistical analysis method detects abnormal pixel distributions that result from LSB modification processes.
- A histogram analysis checks for intensity modifications because it serves to detect hidden information.
- The detection system employs AI algorithms that have undergone training to identify steganographic patterns.

These security threats are addressed through the proposed defensive method that contains:

- Initial data encryption enables the security of message content since it remains protected under any steganography breach.
- The adaptive LSB embedding approach conducts randomised bit modification to minimise statistical abnormalities that would otherwise make discovery easier.
- Steganographic modifications succeed better through the use of images featuring natural patterns with high entropy, such as textual components and noise.
- Steganalysis tools detect minimal distortion through PSNR and MSE optimisation to keep attackers from discovering the hidden information.

The system's defensive methods substantially boost its attack resistance while protecting data integrity.

B. Ethical Implications and Responsible Usage

The valuable capability of steganography for secret communication generates multiple ethical questions about its use. The practice requires evaluation based on these main ethical points:

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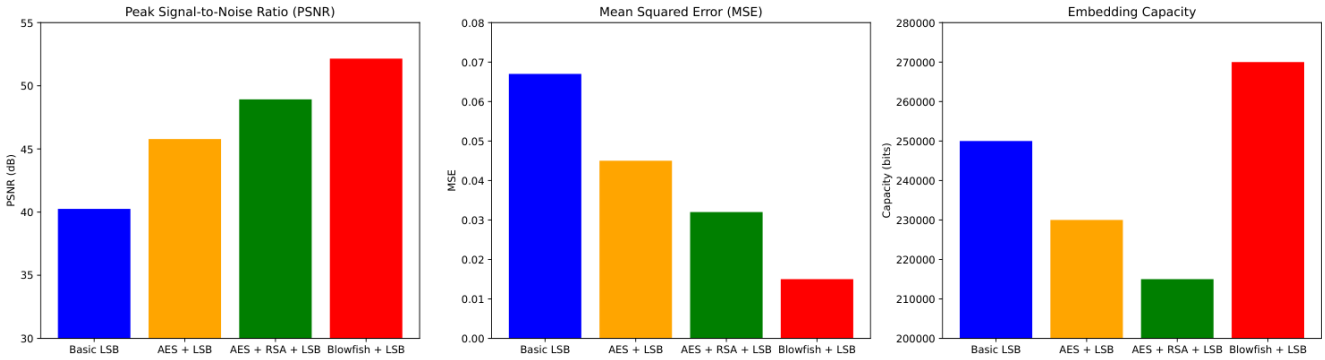


Fig. 2. Comparison of steganographic methods based on PSNR, MSE, and capacity.

- The secure file transmission tool of steganography faces restrictions because it allows unauthorized activities and hiding from officials. The practices must remain within stated privacy laws and should adhere to cybersecurity best practices.
- The organizations must establish ethical regulations defining how steganography applications are used only for secure communication systems, digital watermark security, and data protection needs.
- Users need transparency about ethical factors associated with steganography to avoid unethical uses, yet maintain responsible utilization.

C. *Limitations and Potential Weaknesses*

The proposed method comes with specific drawbacks that need to be considered.

- The encryption system's double process generates an extra computational burden, which affects real-time applications and consumes large amounts of resources.
- Large structured data insertion can result in detectable image distortions that limit the capacity of associated payloads.
- Future steganalysis techniques will depend on advanced machine learning for better pattern detection, which may force continual development of steganographic algorithms to prevent detection.

The difficulty in securing encryption key distribution persists as the failure to protect the encryption key makes hidden data susceptible to exposure. The resolution of forthcoming research will study adaptive encryption technology while developing deep learning anti-steganalysis tools and implementing dynamic key exchange systems.

VI. Conclusion

The research introduces a steganographic solution that combines Blowfish encryption with LSB steganography to augment security and confidential data integrity. The system implements double encryption to protect the encrypted data so that extraction of the steganographic layer cannot compromise its security. The experimental data show increased Peak Signal-to-Noise Ratio (PSNR) values. It reduced Mean Squared Error (MSE) while providing more space for embedding purposes in this method, making it suitable for secure digital communication. Implementing this system contributes to cybersecurity needs, confidential communication systems, and digital forensics investigations. Sensitive data transmission to government agencies, financial institutions, and healthcare providers becomes possible through this method for secure delivery. Digital watermarking demonstrates benefits from this technique, which strengthens copyright protection and attests to genuine data sources. Future enhancements can optimise the effectiveness of the research findings that demonstrate potential. The application of AI-based adaptive steganography enables automatic selection of embedding approaches, improving hidden data's detection difficulty. AI detection methods can be defeated with anti-steganalysis mechanisms that deep learning methods would help develop. When combined with optimised payload compression and security protocols based on quantum encryption, the system achieves better operational speed and extended protection mechanisms. This research demonstrates the necessity of integrating encryption with steganography to achieve secure digital transmission. The developed research creates essential groundwork for AI-empowered steganographic techniques through its treatment of standard weaknesses and introduction of performance improvements. This system maintains its ability to conduct secure, covert data transmission while allowing for expansion through the evolution of digital threats, so researchers continue to develop innovative approaches in cryptography and steganography.

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